Nature Methods

Predicting bacterial community assemblages using an artificial neural network approach

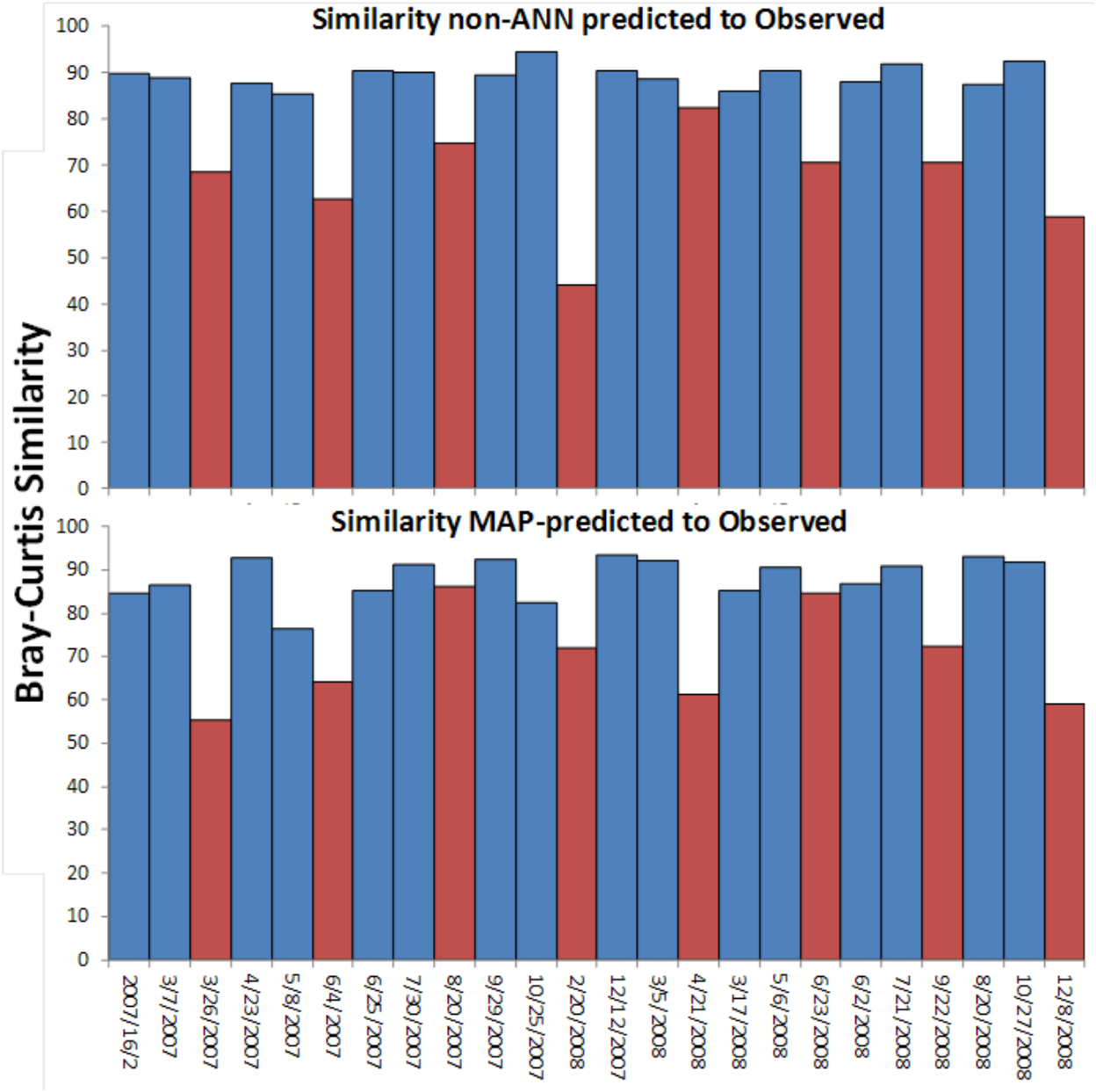
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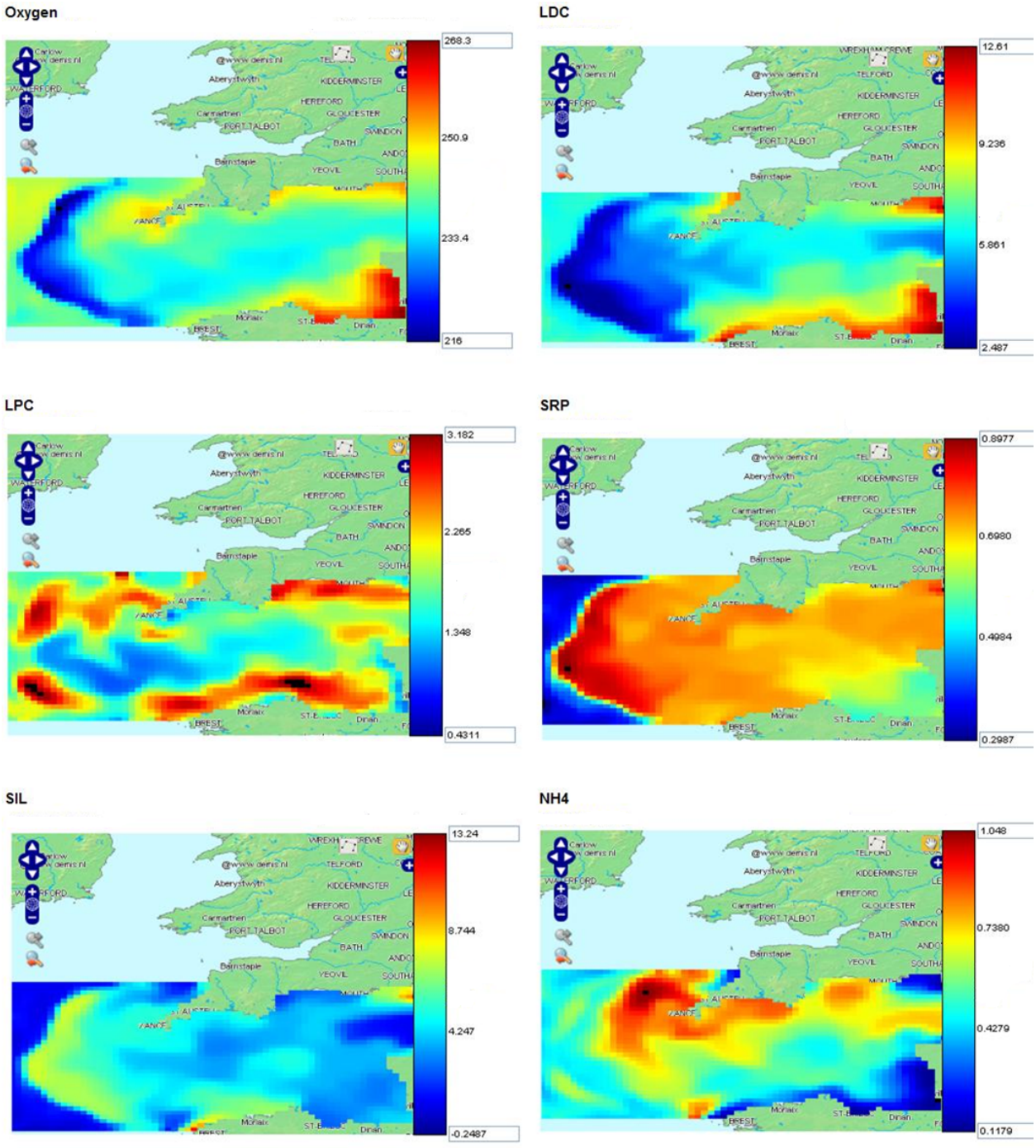
Supplementary Figures

Supplementary Figure 1. Similarity between observed microbial populations and MAP or non-ANN model predictions. Weighted Bray-Curtis similarities between observed microbial population structures and populations predicted by non-ANN model (top) or MAP ANN-based model (bottom). Training data points are blue, validation data points are red.



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Supplementary Figure 2. Region of lowest similarity for predicted microbial populations for December 8, 2008. This region corresponds to areas of relatively decreased dissolved oxygen, and labile dissolved organic carbon and increased silicate and phosphate concentrations.



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Supplementary Tables

Supplemental Table 1. Equations used in SIM non-ANN and MAP models. DO2 = Dissolved O2, SRP = Soluble Reactive Phosphate, NIT = Nitrate, NH4 = Ammonium, SIL = Silicate, CHL = Chlorophyll A, PAR = Photosynthetic Available Radiation, SPC = Small POC, MPC = Medium POC, LPC = Large POC, LDC = Labile DOC, SDC = Semi-labile POC, BAC = Bacteria.

Taxa (Class)	Abrv.	Non-ANN	MAP
Actinomycetales	act	0.17585345 + (9.1188335 + -0.75666732*SIL/(0.37766099 + SDC))/SDC + 1.1028265/(25.001448/(34.789925 - SRP) + 1.2122275*SDC - 74.332596)	oce + 22.226713*bur/(oce*NH4 + PAR - 24.60882)
Alteromonadales	alt	0.24438807 + 1.2987318/DO2 + 0.85241586/SRP + (4.4345937 - 4.3507457/SPC)/(SPC - 0.77406412)	1.54642*oce + 1.1585543/MPC + 0.013399431*MPC*bur
Bacillales	bac	0.39547876 + ((10.772986*LPC - 130.50729)/(18.131842 - PAR) - 182.5119)/(18.155346*SPC - SPC*MPC - 57.199745)	0.20920011 + 0.78153712/(51.184517 - CHL) + (266.07321/(101.37847 - CHL) - 5.6346974)/(1.738047*NIT + CHL - 104.57217)
Burkholderiales	bur	(6.9208775*PAR - 171.74034)/(DO2*SPC - 22.56266*SPC) + 25.779634/PAR - 0.020171262	0.25217551 + 0.23017572*pse*pse - 0.12012058/(1.5346673 - 1.8423156*des - 0.19410059*pse)
Cyanobacteria	cya	1.5356853 + 158.05879/(1380.1102 - 113.52555*BAC) + (0.88573962*SPC - 67.378265)/(LDC - 44.242058)	1.3684113 + 1.9893361*pro + 0.38945326/(1.3701524*pse - 2.674154)

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Desulfobacterales	des	1.1616918 - 32.811405/((NH4 - 0.28713077*SIL)/(SRP - 12.060976) - 89.543625/(NH4 - SIL) - SIL)	1.0148802*leg/(0.028500969*SDC + leg - 0.034206811/(0.39122796 - pla) - 0.0027609211)
Enterobacteriales	ent	0.062237434 + 0.0068561183*NIT + 0.0029438282/(SDC - 1.4696478) + 3.1026642/(SDC - NIT - 3.6096344)	0.1857235 + 0.0064113638/(2.4138918 - alt) + 0.63271379*pla + 0.1382945*rhi*rhi - 0.45226917*rhi
Flavobacteriales	fla	3.9980876 + 1.6052152/(2.0392702 - SIL) + 0.00033707957*SIL*PAR + - 0.02931348*PAR/(SRP - 58.893883) - 8.6363077/PAR	(423.85098 + PAR - 1.5280328*NH4 - 419.50925/(NH4 - PAR - 7.8205767) - 45.470375*opi)/(92.979614 + (1.3571365*NH4 - PAR)/(CHL - 50.180706))
Legionellales	leg	0.14604749 + (0.025784252*PAR - 0.34850338)/(SDC*SDC) - 0.00150406*PAR	(4.9094629 - ric)/(SPC + pro) + 0.050169565*ric - 0.15212995
Methylophilales	met	1.2159082 + 0.30510926/SDC + 2.3284702/(51.131306 - SDC) - 0.96360439/SIL	3.7856104 + (BAC - 0.1529267*PAR)/(0.33591437*BAC* PAR - 4.8799362*PAR) - 20.708906/(4.8110352*ric) - 0.43707278*ric
Oceanospirillales	oce	(1.0208081*NIT + 158.87546/LDC)/(35.850037 + (NIT*MPC - 8.2157288*MPC)/BAC) - 0.11653984	(1.0208081*NIT + 158.87546/LDC)/(35.850037 + (NIT*MPC - 8.2157288*MPC)/BAC) - 0.11653984
Opitutales	opi	19.99954/(8.5324106 + MPC*LPC) + 22.744991/(8.5324106 + SRP) - 0.20080672	0.22287618 - 0.046179768/(0.26841781 + bur - 1.3516418/(0.42526227 + alt))
Planctomycetales	pla	1.4422259/((DO2 + 125.2028/DO2)/BAC + 38.180267/NH4 - 1.6605625)	1.4422259/((DO2 + 125.2028/DO2)/BAC + 38.180267/NH4 - 1.6605625)

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Prochlorales	pro	0.16559529 + (388.99155/(PAR - DO2) + 0.56381392*DO2 - 24.580278)/(PAR + LPC/(23.081535 + 232.52052/(PAR - DO2)))	0.16559529 + (388.99155/(PAR - DO2) + 0.56381392*DO2 - 24.580278)/(PAR + LPC/(23.081535 + 232.52052/(PAR - DO2)))
Pseudomonadales	pse	0.43085498 + 0.015779173*LPC + BAC/(1.2447904*LDC*NH4 - 74.767914*LDC) + (0.22288856*LPC - 18.382013)/(LPC + LDC - 95.812164)	1.2520498 + (0.092745684*pla + 0.92952281*des*des/(pla - 0.11784284) - pla*des)/(0.24709482 + des - pla)
Rhizobiales	rhi	0.50823617/(BAC - 12.941306) + 0.0031962304*SDC + 0.0074944119*MPC - 0.00047343565*DO2	0.50823617/(BAC - 12.941306) + 0.0031962304*SDC + 0.0074944119*MPC - 0.00047343565*DO2
Rhodobacterales	rho	2.4198499 + (0.048865978*DO2*PAR - 0.4693734*DO2)/(PAR + 1.048866*LPC)	5.2531962 + (162.51891/LPC - 39.345848*ver)/BAC - 30.418375/PAR
Rhodospirillales	rhs	0.9589597 + 0.0061694593*NH4 + (3.1066315 + 0.23390785*CHL - 0.38064894*LPC)/(1.9566929* LPC - CHL) - 0.55760747/NIT	1.4732952 + 0.0052151866/(0.74428695 - 0.033404239*LDC) + 0.02345931*LDC*opi - 0.034257345/alt - 0.77242297*opi - 0.014427044*LDC
Rickettsiales	ric	5.9613948 + 182.79601/(PAR + DO2 - LPC - 180.38664) + (15.678667 + 2081.6321/(DO2 - LPC - 11.450574))/(DO2*NIT)	2.9441545 + bac + 2.6896143*spb + -6.3412662*spb*spb/(MPC - 4.834919*bac - 2.5394886) - 1.1484582*spb*spb*spb

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Sphingobacteriales	spb	0.49005336 + 0.00083933183/(LDC - 2.8115487) - 1.1127042/((20.494658 + LDC)/(2.6020534 + MPC) + 0.20266914*MPC - SRP) - 0.0038352001*CHL	0.49005336 + 0.00083933183/(LDC - 2.8115487) - 1.1127042/((20.494658 + LDC)/(2.6020534 + MPC) + 0.20266914*MPC - SRP) - 0.0038352001*CHL
Sphingomonadales	sph	0.12129939 + (0.189785 - 0.39707646*SDC)/(1.4661274* SRP - SRP*SDC)	(1.0146122 + (8.8441858*bur + 0.33803105*bur*DO2)/(17.901848 - 5.2209659*pse))/(24.813274 + 32.444782*bur - 19.686251*pse) - 0.0071777287
Verrucomicrobiales	ver	0.95215714 + 0.0032644426*BAC + 0.714706/(DO2 - 48.445137) - 7.1619361e-5*DO2*DO2 - 0.0031083329*CHL	0.95215714 + 0.0032644426*BAC + 0.714706/(DO2 - 48.445137) - 7.1619361e-5*DO2*DO2 - 0.0031083329*CHL
Vibrionales	vib	0.035799664 + (0.51420009*PAR/SPC - 0.67304206)/(NH4 - 48.320122)	0.073676355 + 19.065651*leg*oce*spb*spb*spb - 0.96004277*leg*spb*spb
Xanthomonadales	xan	0.012212357/(SPC - 3.2806535) + 0.0025860486*MPC/SIL - 0.0011210864	0.020732749*vib + 0.016223148*leg*leg/(leg*vib - 0.063814372)

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Supplemental Table 2. Describing the functional parameters that provide the best prediction of relative abundance of each taxon. 'Abrv.' are abbreviations for taxa used in equations. 'Function Parameters' are specific taxa or environmental parameters used in function to determine relative abundance of taxa. DO2 = Dissolved O_2, SRP = Phosphate, NIT = Nitrate, NH4 = Ammonium, SIL = Silicate, CHL = Chlorophyll A, PAR = Photosynthetic Available Radiation, SPC = Small POC, MPC = Medium POC, LPC = Large POC, LDC = Labile DOC, SDC = Semi-labile POC, BAC = Bacteria, TMP = Temperature.

Taxa (Class)	Abrv.	*In situ*-based MAP Function Parameters	SIM-based MAP Function Parameters
Actinomycetales	act	sgm, bac, TMP	oce, bur, NH4, PAR
Alteromonadales	alt	PAR, SAL, NH4, SIL	oce, MPC, bur
Bacillales	bac	TMP, NH4	CHL, NIT, CHL
Burkholderiales	bur	PAR, SIL	pse, des
Cyanobacteria	cya	NO2, PAR, SIL, rhs	pro, pse
Desulfobacterales	des	NO3, PAR, NO2	leg, SDC, pla
Enterobacteriales	ent	bur, pla, SIL, NO3	alt, pla, rhi
Flavobacteriales	fla	PAR, CHL, TMP, NO3, NO2	PAR, NH4, opi, NH4, CHL
Legionellales	leg	NO2, PAR, SIL, SRP	ric, SPC, pro
Methylophilales	met	TMP, NH4, NO3, SIL	BAC, PAR, ric
Oceanospirillales	oce	PAR, CHL, bac	NIT, LDC, MPC, BAC
Opitutales	opi	NO2	bur, alt
Planctomycetales	pla	CHL, PAR, NH4	DO2, BAC, NH4
Prochlorales	pro	NO3, PAR, NH4, SAL	PAR, DO2, LPC
Pseudomonadales	pse	NH4, PAR, NO3	pla, des
Rhizobiales	rhi	CHL, PAR	BAC, SDC, MPC, DO2
Rhodobacterales	rho	PAR, CHL	LPC, ver, BAC, PAR
Rhodospirillales	rhs	NO3, SIL, TMP, NH4, SAL, CHL, SRP	LDC, opi, alt
Rickettsiales	ric	SIL, TMP, NH4, SAL, CHL, PAR	bac, spb, MPC
Sphingobacteriales	spb	PAR, SIL	LDC, MPC, SRP, CHL
Sphingomonadales	sph	NH4, PAR, NO3, NH4, SIL, TMP	bur, DO2, pse, bur
Verrucomicrobiales	ver	TMP, NH4, PAR	BAC, DO2, CHL
Vibrionales	vib	SIL, SAL, SRP	leg, oce, spb
Xanthomonadales	xan	act, leg, NO3	vib, leg

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Supplemental Table 3 – The list of 16 environmental variables used. Six are similar, but not identical, in both the SIM and *in situ* data sets, seven are SIM-specific and three are *in situ* data-specific.

Count	SIM Variables	*in situ* variables
1	dissolved O_2 (mmol $m^{-3)}$	n/a
2	Soluble Reactive Phosphate (mmol/m^3)	soluble reactive phosphate (µmol L-1)
3	nitrate (mmol/m^3)	nitrate (µmol L-1)
4	ammonium (mmol/m^3)	ammonia (µmol L-1)
5	silicate (mmol/m^3)	silicate (µmol L-1)
6	chlorophyll A (mg C/m^3)	chlorophyll A (µg/L)
7	photosynthetically active radiation (PAR, W/m^2)	photosynthetically active radiation (PAR) (hrs daylight)
8	small size particulate organic carbon (sPOC) (mg C/m^3)	n/a
9	medium (m)POC (mg C/m^3)	n/a
10	large (l)POC (mg C/m^3)	n/a
11	labile dissolved organic carbon (LDOC) (mg C/m^3)	n/a
12	total bacterial carbon (mg C/m^3)	n/a
13	semi-labile (SL)-DOC (mg C/m^3)	n/a
14	n/a	salinity (PSU)
15	n/a	NO2 + NO3 (µmol L-1)
16	n/a	temperature (ºC)

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Supplementary Results

MAP model fits observed data

For *in situ* model, the highest similarity occurred on April 11th, 2006 (96.9) and the lowest on August 4th, 2003 (87.5). This model was then used to predict the relative abundance of the 24 bacterial Orders in the 24 validation time points. The average weighted Bray-Curtis similarity between the observed and predicted data for these 24 time points was 88.6 (p value 0.00025). The highest similarity occurred on July 30th, 2008 (similarity 92.0), while the lowest occurred on December 12th 2007 (similarity 81.6).

For SIM model, the highest similarity in the model-training data occurred on August 9th, 2007 (similarity 92.4) and the lowest on May 8th 2007 (similarity 76.4). This model was then used to predict the relative abundance of the 24 bacterial Orders in the 8 validation time points. The average weighted Bray-Curtis similarity between the observed and predicted data for these 8 time points was significant (73.9; p-value 0.00159). The highest similarity occurred on September 29th, 2007 (85.0), while the lowest occurred on February 20th, 2008 (61.0).

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